

Recursive Offline Distillation via Epistemic Complexity Approximation

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Abstract

Distillation of reasoning chains from Large Language Models (LLMs) into their smaller siblings is a standard training approach. Frequently researchers do not have full access to a teacher model and its internal representations, which limits the resulting performance. We propose a novel blueprint for efficient offline distillation that recursively uses lightweight epistemic and aleatoric complexity approximations for selective distillation. Specifically, for three open models (3-4B) at each epoch we estimate epistemic uncertainty using a medium model (70-72B) as an aleatoric proxy and sample only a small subset of the reasoning chains for distillation. Our approach consistently outperforms naive distillation on comparable computational budget (57.15 ± 1.01 vs 51.61 on average) and offers a much steeper learning curve. Additionally, we present a comparative analysis of methods to generate and normalize distillation traces. We publish our code and data¹.

1 Introduction

Modern LLMs come in different sizes, and their performance is often correlated with the number of parameters. The largest of them demonstrate exceptional results and span 1-2 trillion parameters. Further domain adaptation (Cheng et al., 2024; Lu et al., 2024) or even inference at this scale requires massive computational resources (Wan et al., 2024) unavailable to most practitioners. It prompts the need for capable smaller models (Chen and Varoquaux, 2026) that can be fine-tuned and deployed in resource-constrained environments.

A prominent approach for small model training is distillation from a larger teacher model (Hinton et al., 2015; Sanh et al., 2020; Xu et al., 2024). Many methods rely on having access to the online teacher (Agarwal et al., 2024; Jiao et al., 2020; Gu et al., 2024), which is not always possible (Zhou

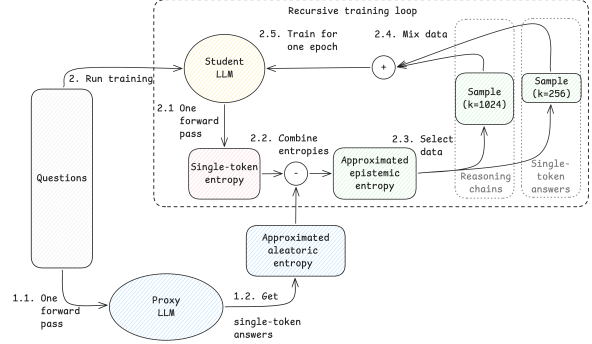


Figure 1: Recursive offline distillation scheme

et al., 2023; Ryskulov et al., 2026). It forces practitioners to explore offline distillation, where the student model is trained on the pre-generated teacher’s text output (Li et al., 2023; Hsieh et al., 2023).

We propose a fully automated recursive pipeline for offline distillation (Figure 1, described in 3) based on uncertainty decomposition (Wang et al., 2025b). The method consistently outperforms naive distillation (57.15 ± 1.01 vs 51.61 on average), and it is designed to work in resource-constrained environments.

Additionally, we provide a comparative analysis of augmentation strategies for synthetic reasoning chains when the teacher model does not perform perfectly on the target distillation domain and conduct sensitivity studies of the synthetic trace generation and normalization methods and sampling strategies.

2 Related Work

Recent work improved distillation through training objectives (Ko et al., 2024; Xu et al., 2026), data selection (Just et al., 2026; Li et al., 2025; Yang et al., 2026), and curriculum learning (Jiang et al., 2025; Zhang and Liu, 2026). Joint supervision with rationales and labels improves data efficiency (Hsieh et al., 2023). Curriculum benefits vary with model capability, task complexity, and the choice

¹TODO

of difficulty metric (Jia et al., 2026). ELAD (Zhang et al., 2024) selects uncertain student explanations for teacher revision. Complexity-aware fine-tuning (Goncharov et al., 2026) selects reasoning examples by single-token entropy. We recursively sample pre-generated traces using student-proxy entropy differences.

Trace synthesis includes explaining known solutions (Li and Mooney, 2024; Wang et al., 2025a) and learning from corrections (An et al., 2024; Yang et al., 2025). We compare direct generation and various augmentations on the same task under constrained trace lengths.

3 Method

Our central hypothesis is that we can approximate epistemic uncertainty of the student model with its entropy and entropy of a larger LLM. Goncharov et al. (2026) show that a single-token entropy (Figure 4) is a reasonable complexity predictor for a given entry in a question-answering dataset. We assume that it approximates the total uncertainty and that we can decompose it into epistemic and aleatoric components (Depeweg et al., 2018; Malinin and Gales, 2021). With scaling laws (Kaplan et al., 2020) in mind, we hypothesize that larger models should have lower total uncertainty. Therefore, in comparison to a small student model (3-4B) its substantially larger sibling (70-72B²) could be treated as a weak aleatoric proxy.

For a question x , assume $H(x) = A(x) + E(x)$, where H_S , H_P - single-token entropy (student and proxy), A - shared aleatoric component, and E_S , E_P - model-specific epistemic component. If proxy’s residual epistemic uncertainty satisfies $E_P(x) \leq \varepsilon(x)$, then

$$\begin{aligned} A(x) &= H_P(x) - E_P(x) \geq H_P(x) - \varepsilon(x) \\ E_S(x) &= H_S(x) - A(x) \leq H_S(x) - H_P(x) + \varepsilon(x) \end{aligned} \quad (1)$$

Thus, the entropy difference (*entropy gain*) approximates the student’s epistemic uncertainty with error at most $\varepsilon(x)$.

The proposed pipeline consists of the following steps (Figure 1, Algorithm 1): (1) prompt the larger model to answer all questions in the dataset with a single token once (or an ensemble of models taking average), collect entropy, assume it is an aleatoric uncertainty proxy; (2) for each training

Algorithm 1 Recursive offline distillation via epistemic complexity approximation

Require: Student S , proxy model P , epochs T
Require: Questions D , ground-truth answers y_x , and pre-computed reasoning traces r_x
Ensure: Trained student S

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1: for all  $x \in D$  do
2:    $H_P(x) \leftarrow \text{ENTROPY}(P, x)$ 
3: end for
4: for  $t = 1, \dots, T$  do
5:   for all  $x \in D$  do
6:      $H_S(x) \leftarrow \text{ENTROPY}(S, x)$ 
7:      $\text{EG}(x) \leftarrow \max\{H_S(x) - H_P(x), 0\}$ 
8:   end for
9:    $Q \leftarrow \text{WEIGHTEDSAMPLE}(D, \text{EG}, 1024)$ 
10:   $Q_A \leftarrow \text{first 256 questions in } Q$ 
11:   $B_R \leftarrow [(x, r_x) : x \in Q]$ 
12:   $B_A \leftarrow [(x, y_x) : x \in Q_A]$ 
13:   $S \leftarrow \text{TRAIN}(S, B_R \parallel B_A)$ 
14: end for
15: return  $S$ 

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epoch, (2.1) prompt the student model to answer all questions in the dataset with a single token, collect entropy; (2.2) calculate entropy gain $\text{EG}(x) = \max\{H_S(x) - H_P(x), 0\}$; (2.3.1) sample 1024 synthetic reasoning chains without replacement, weighted by positive entropy gain; (2.3.2) sample 256 single-token answers in similar fashion as a subset of the previous sample; (2.4) concatenate samples; (2.5) train the student using LoRA (Hu et al., 2021) on reasoning traces first, followed by single-token responses.

4 Results

4.1 Experimental Setup

Dataset: Multiple choice question answering dataset MMLU Pro (Wang et al., 2024). It has questions of different complexities from 14 domains, ten options for each, single correct one. Random train/test split - 80/20.

LLMs: Aleatoric proxy: ensemble of two medium models³ (taking average of their entropies) - LLaMA 3.3 70B (Grattafiori et al., 2024), Qwen 2.5 72B (Qwen-Team, 2024). Students: three small models Qwen2.5 3B (Qwen-Team, 2024), Phi-4-Mini (Microsoft et al., 2025), LLaMA 3.2 3B (Grattafiori et al., 2024). Reasoning trace source: DeepSeek V4 Pro and Flash (DeepSeek-AI et al., 2026) via API.

²This particular size is chosen based on the imposed resource constraints (80GB VRAM) and open model availability

³All models and datasets are published under permissive licenses that allow them to be used for research purposes

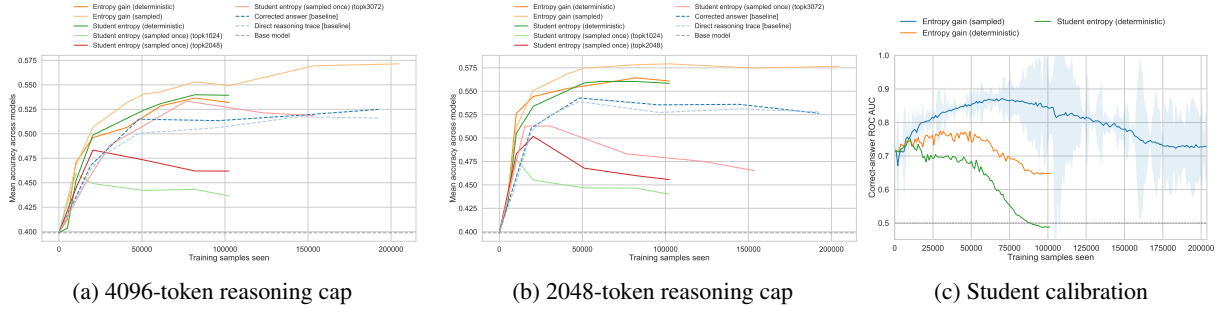


Figure 2: Distillation pipeline results across three models (average) on MMLU-Pro

Method	LLaMA-3B	Qwen-3B	Phi4-mini	Mean
Direct reasoning trace	46.09 / 45.55	51.45 / 50.37	57.27 / 62.43	51.61 / 52.78
Corrected answer	47.80 / 47.26	50.87 / 49.42	<u>58.81</u> / 61.26	52.49 / 52.65
Student entropy (deterministic)	49.83 / 50.08	53.62 / 53.03	<u>58.35</u> / 64.42	53.93 / 55.85
Entropy gain (deterministic)	<u>50.17</u> / <u>50.37</u>	51.04 / <u>53.28</u>	58.40 / 64.63	53.20 / <u>56.10</u>
Student entropy (sampled once, $k = 1024$)	41.35 / 40.11	39.44 / 39.07	50.17 / 52.87	43.65 / 44.01
Student entropy (sampled once, $k = 2048$)	42.19 / 41.06	43.18 / 43.02	53.20 / 52.62	46.19 / 45.57
Student entropy (sampled once, $k = 3072$)	42.02 / 42.81	56.77 / 40.86	56.77 / 55.90	51.86 / 46.52
Entropy gain (sampled, ours)	52.58 $\pm$ 1.94 / 52.08 $\pm$ 2.28	<u>56.64</u> $\pm$ 3.79 / 56.86 $\pm$ 2.32	62.22 $\pm$ 2.42 / 64.07 $\pm$ 4.52	57.15 $\pm$ 1.01 / 57.67 $\pm$ 1.19

Table 1: Distillation pipeline last epoch accuracy (4096-token / 2,048-token thinking budget) on MMLU-Pro.

Hardware: We run all experiments⁴ on a single A100/H100⁵ GPU with 80GB of VRAM to validate the method in resource-constrained environments.

4.1.1 Our method and results

We evaluate the proposed pipeline for 200 epochs (205k questions seen) and 2 thinking budgets (2048 and 4096 tokens). It is compared at similar data budget with:

Direct reasoning trace, Corrected answer - naive distillation on the most promising types of reasoning traces (see 4.3);

Student entropy (deterministic), Entropy gain (deterministic) - alternative pipeline with weighted sampling replaced with greedy sampling;

Student entropy (sampled once, $k=x$) - distillation executed against k samples selected once by student entropy (not recursively at each epoch).

Entropy gain (sampled), Student entropy (deterministic), Entropy gain (deterministic) use the proposed data mix (reasoning traces/single-token answers = 1024/256), other methods use only reasoning traces.

Notably, the proposed data mix increases the number of training tokens only by 0.003% - 256 extra tokens compared to 8,388,608 tokens in 1024 reasoning chains.

All randomly sampled methods use 3 random

seeds, reporting $\pm 95\%$ confidence-intervals, and deterministic methods are evaluated once with greedy sampling.

4.2 Main Results

Table 1 and Figure 2 show the results for MMLU Pro. We see that the proposed method produces a significant improvement over naive distillation (up to 10% gain). For both reasoning budgets, it provides the best result on average across three models in comparison with other approaches. Greedy sampling demonstrates subpar and less stable curves, though surpassing the baselines. Greedy distillation based on student entropy sampled once (Goncharov et al., 2026) shows signs of severe degradation after the initial uplift. However, Qwen 2.5 3B, notably, demonstrates the best performance with it at $k=3072$, but degrades at smaller sample sizes, revealing the fragility of the approach.

We also note that our method helps with the model calibration significantly enhancing ROC AUC of single-token entropy for the first half of training. Shortly after, it starts declining which is loosely correlated with the method’s plateau.

4.3 Synthetic Traces Generation

In resource-constrained environments, available VRAM imposes a strict limitation on the length of distilled reasoning chains. As soon we start truncating them, we face a new challenge - which part

⁴Training hyperparameters can be found in Appendix A.

⁵We had to switch cards depending on availability

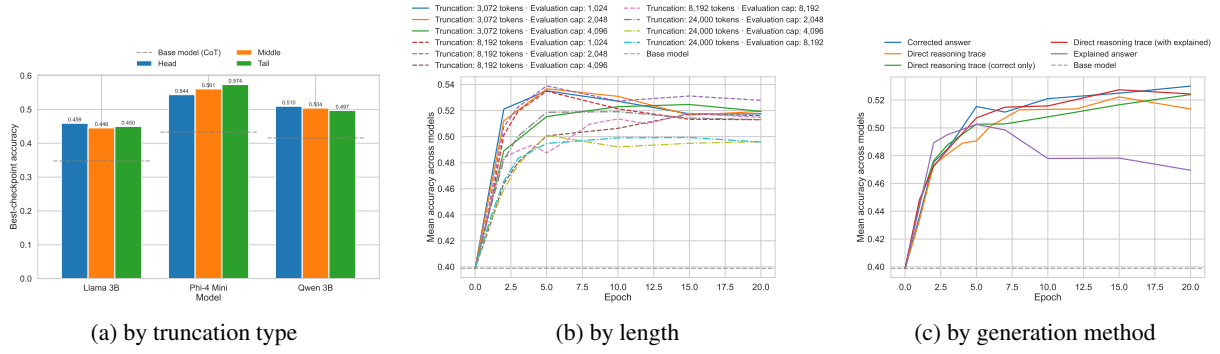


Figure 3: Comparison of synthetic traces across three models (average)

of the trace contains the most useful signal? At last, how exactly do we prompt the teacher model to generate reasoning traces?

We evaluate reasoning traces by: (a) *truncation type* - which part of the trace we keep when it exceeds the maximum length; (b) *length* - what is the optimal training truncation limit and evaluation thinking budget (in tokens). (c) *generation method* (below).

Generation methods (details in Appendix C): (1) *Direct reasoning trace* - baseline prompt to just answer the question; (2) *Direct reasoning trace (correct only)* - same as (1), but we keep only traces with correct answers; (3) *Explained answer* - we give a correct answer to the model and prompt it to explain it; (4) *Direct reasoning trace (with explained)* - same as (1), but replace incorrect traces with (3); (5) *Corrected answer* - same as (1), but for incorrect traces we give the model correct answers and ask it to update its reasoning.

Figure 3 shows that models have a slight preference towards head truncation, medium to short reasoning chains in training and thinking budgets. Individual model curves are provided in Appendix C.3. Surprisingly, longer reasoning chains and greater thinking budgets lead to notable performance decrease. As to the generation methods, we see that augmented reasoning traces lead to elevated performance, while the consistent data leakage results in degradation. Based on the findings, we use 8,192-token head truncation and *Corrected answer* generation for the pipeline.

4.4 Sensitivity Study

To attribute performance gains to either the data mix or the entropy-based score and sampler, we compare the proposed pipeline (*entropy gain*) with: (1) alternative pipelines where we sample according to *student entropy* and *discounted entropy gain*

Sampler	Three-model mean
Direct reasoning trace (baseline)	51.61
Random	55.08 $\pm$ 0.80
Student entropy	55.87 $\pm$ 1.40
Discounted entropy gain	55.98
Entropy gain	57.15 $\pm$ 1.01

Table 2: Distillation pipeline last epoch accuracy for different sampling methods (4096-token thinking budget)

(see below); (2) alternative pipeline where we use a random sampler.

Discounted entropy gain is defined as $\max\{H_S(x) - k(t) * H_P(x), 0\}$, $k(t) \in [0, 1]$, where t - training epoch. We set $k(t)$ initially to 0.8 and decrease it according to a schedule, hypothesizing that as the student undergoes training the static medium model stops being a good aleatoric proxy.

Table 2 shows that across three models on average entropy gain provides the best results confirming the contribution of the aleatoric uncertainty approximation. At the same time, we notice that random sampler performs exceptionally well (confirming findings of Okanovic et al. (2023)) compared to the baseline, which points to the proposed data mix as a significant contributor to the gains.

5 Conclusion and Discussion

This paper introduces a recursive offline distillation method rooted in epistemic uncertainty approximation (57.15 $\pm$ 1.01 mean accuracy across three students on MMLU-Pro vs 51.61 for direct distillation, offers a much steeper learning curve (baseline score at 15% of its data budget)). We confirm that both the entropy difference sampler and the data mix contribute to the gains. Surprisingly, shorter traces and reasoning budgets perform better. Further exploration of the more rigorous epistemic uncertainty estimation and the surprising effect of the

small data mix augmentation remain future work.

Limitations

- Method is tested only with small models. We do not know how well it is going to generalize to larger LLMs. Additionally, it might be challenging to find a suitable aleatoric proxy for larger students.
- We use a single dataset. We do not know how well it translates to other settings.
- Alternative aleatoric approximations, such as agreement of the ensemble and other, are left unexplored.
- We did not test and compare multiple sizes of aleatoric proxies and their impact on performance.
- Different compositions of the data mix are not tested due to lack of resources.
- We tested only a single schedule for *Discounted Entropy Gain*. Alternatives are yet to explore.
- Some low-resource environments might have even more constraints and therefore struggle to fit a suitable aleatoric proxy into VRAM.

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A Experimental Hyperparameters

Hyperparameter	Value
Optimizer	AdamW (adamw_torch)
Learning rate	10^{-4}
Learning-rate schedule	Cosine
Warmup ratio	0.06
Weight decay	0
Adam (β_1, β_2)	(0.9, 0.999)
Adam epsilon	10^{-8}
Maximum gradient norm	1.0
Mixed precision	BF16 enabled
Gradient checkpointing	Enabled
torch.compile	Enabled
LoRA rank r	8
LoRA alpha	16
LoRA scaling α/r	2
LoRA dropout	0.1
Initialization	Gaussian

B Single-token entropy

Single-token entropy (Figure 4) measures a model’s uncertainty over the first answer token

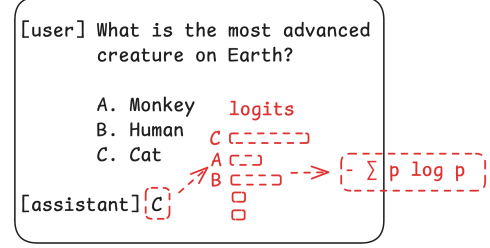


Figure 4: Single-token entropy example

when prompted to answer a question with a single token. For a question x , we apply softmax to the first-token logits to obtain probabilities $p(v | x)$ over the model’s full vocabulary V , and compute normalized Shannon entropy:

$$H(x) = -\frac{1}{\log |V|} \sum_{v \in V} p(v | x) \log p(v | x). \quad (2)$$

Dividing by $\log |V|$ puts student and proxy entropies on a common $[0, 1]$ scale despite differences in vocabulary size: values near zero indicate a concentrated prediction, while values near one indicate a nearly uniform distribution.

C Synthetic trace generation methods

We compare five ways to construct reasoning-and-answer targets for the same multiple-choice questions. Directly generated option letters are checked against ground-truth labels.

Direct reasoning trace. The teacher receives the question and options, generates reasoning, and predicts an option without seeing the ground-truth answer. The baseline retains both correct and incorrect responses.

Direct reasoning trace (correct only). We retain only baseline responses whose predicted option matches the ground truth, reducing the training set by discarding failed attempts.

Explained answer. The teacher receives the question, options, and correct answer. It generates a step-by-step explanation that reads as a fresh solution, allowing alternatives and self-correction while avoiding acknowledgment of the supplied answer. The target pairs this explanation with the ground-truth option.

Direct reasoning trace (with explained). We retain correct baseline responses and replace each

incorrect response with the explained-answer response for the same question. This preserves question coverage while restricting answer-conditioned generation to failed attempts.

Corrected answer. For each incorrect baseline response, the teacher receives the original reasoning, its incorrect option, and the correct option. It continues the trace toward the correct answer, presenting a natural self-correction without restarting or repeating the original text. We append the continuation and use the ground-truth option as the final answer. Correct baseline responses remain intact.

C.1 Prompt templates

The placeholders below are filled with the subject, question, and options. Correct-only filtering and replacement with explanations reuse these prompts; they introduce no additional generation prompt.

Direct generation: system prompt. Reasoning is collected separately from the teacher’s final answer.

The following are multiple choice questions about {subject}. Choose a correct option letter. Answer with a single symbol. Do not print anything else.

Explained answer: system prompt.

The following is a multiple choice question about {subject}. The correct answer is provided to you, but your task is to produce a detailed chain-of-thought with step-by-step reasoning that reads as if you were solving the question from scratch—including any natural uncertainty, consideration of multiple options, or self-correction. Do not state or imply that the answer was given to you. End your response with Answer: <letter>.

Corrected answer: system prompt.

The following are multiple choice questions about {subject}. You will be shown a question, a partial chain-of-thought that was previously produced for it, the incorrect option that chain-of-thought arrived at, and the correct option letter. Continue the partial chain-of-thought from where it left off so that it naturally arrives at the correct option. Do not restart the reasoning, do not repeat what was already written, and do not acknowledge that the previous attempt was wrong or that the correct answer was given to you. Write the continuation as if you were the same reasoner noticing a mistake or new consideration mid-thought and revising course. In the end, answer with the correct option letter. End your response with Answer: <letter>.

User prompts. All three begin with:

Question: {question}
Options:
a. {option text}
b. {option text}
...

For explained answers, append Correct option: followed by the ground-truth letter. For corrections, append:

Partial reasoning so far:
{original reasoning}
Incorrect option this reasoning led to:
{predicted letter}
Correct option: {ground-truth letter}

C.2 Illustrative examples

For brevity, consider a toy question with two options: “A rectangle has sides of length 3 and 2. What is its perimeter? a. 6; b. 10.” The following illustrative targets show how the methods handle an incorrect attempt.

Direct trace. “The perimeter is $3 \times 2 = 6$. Answer: a.” The baseline retains this response; the correct-only variant discards it.

Explained answer. Given correct option b, a fresh explanation is: “There are two sides of each length, so the perimeter is $2(3 + 2) = 10$. Answer: b.” The with-explained variant substitutes this response for the incorrect direct trace.

Corrected answer. Given the original reasoning, incorrect option a, and correct option b, a continuation is: “That product gives the area. For the perimeter, I need to add all four sides: $3 + 2 + 3 + 2 = 10$. Answer: b.” The target retains the original reasoning, appends this continuation, and ends with b.

C.3 Per-model results

Figures 5–7 show test accuracy across training epochs for each student. Length comparisons use direct reasoning traces; generation-method comparisons use head truncation at 8,192. Dashed horizontal lines and epoch-zero values use the base model’s accuracy prompted to answer with chain-of-thought (base models originally are not reasoners), shared across evaluation budgets.

D Sensitivity Study

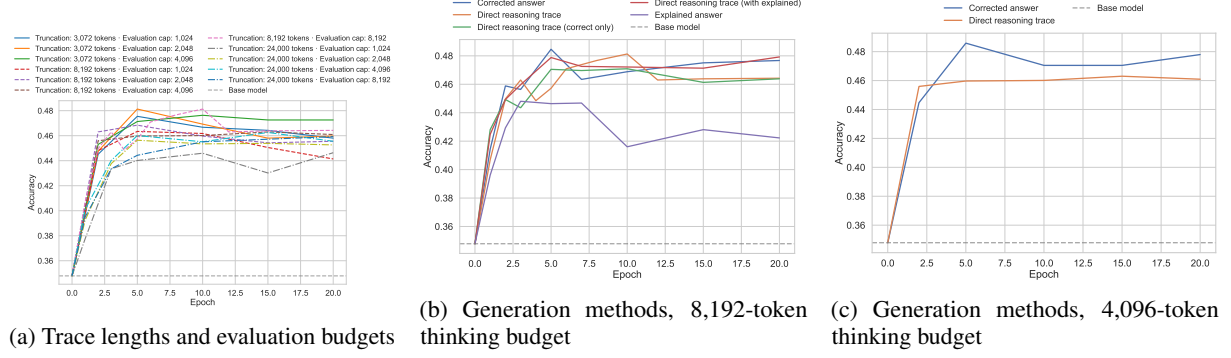


Figure 5: Synthetic-trace distillation results for LLaMA-3B.

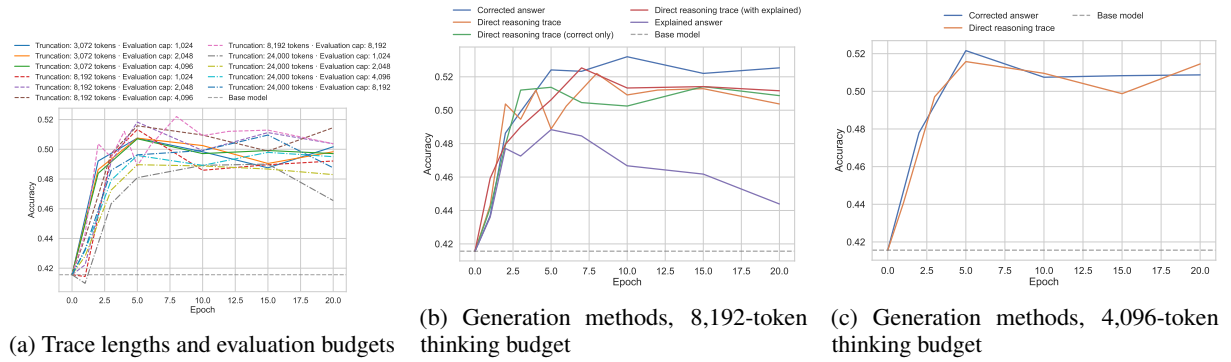


Figure 6: Synthetic-trace distillation results for Qwen-3B.

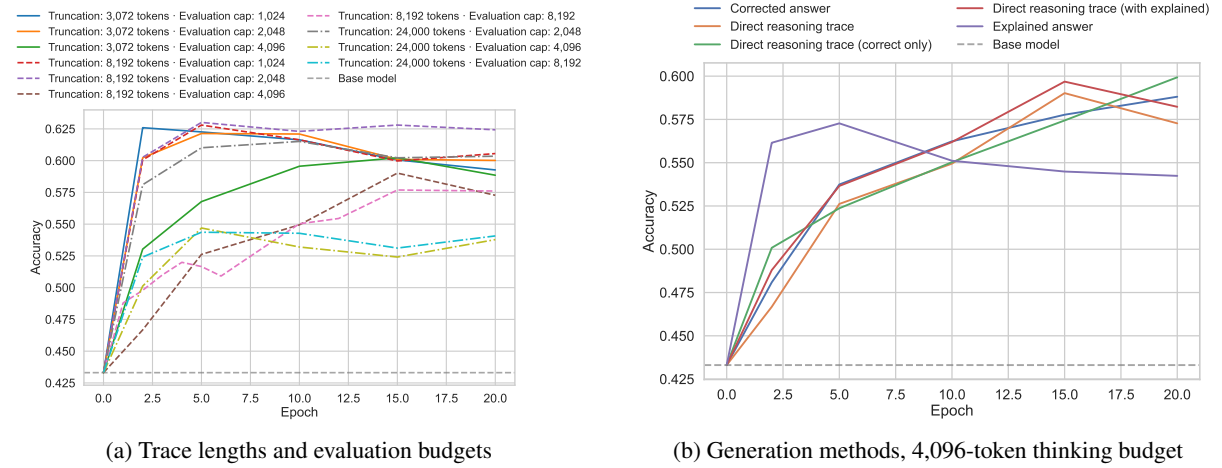


Figure 7: Synthetic-trace distillation results for Phi4-mini.

Sampler	LLaMA-3B	Qwen-3B	Phi4-mini	Mean
Direct reasoning trace (baseline)	46.09	51.45	57.27	51.61
Random	51.77 ± 2.76	57.91 ± 1.40	55.57 ± 2.54	55.08 ± 0.80
Student entropy	51.18 ± 3.11	57.32 ± 1.52	59.12 ± 5.22	55.87 ± 1.40
Discounted entropy gain	51.50	56.90	59.53 ± 2.11	55.98
Entropy gain	52.58 ± 1.94	56.64 ± 3.79	62.22 ± 2.42	57.15 ± 1.01

Table 3: Distillation pipeline last epoch accuracy for different sampling methods (4096-token thinking budget)